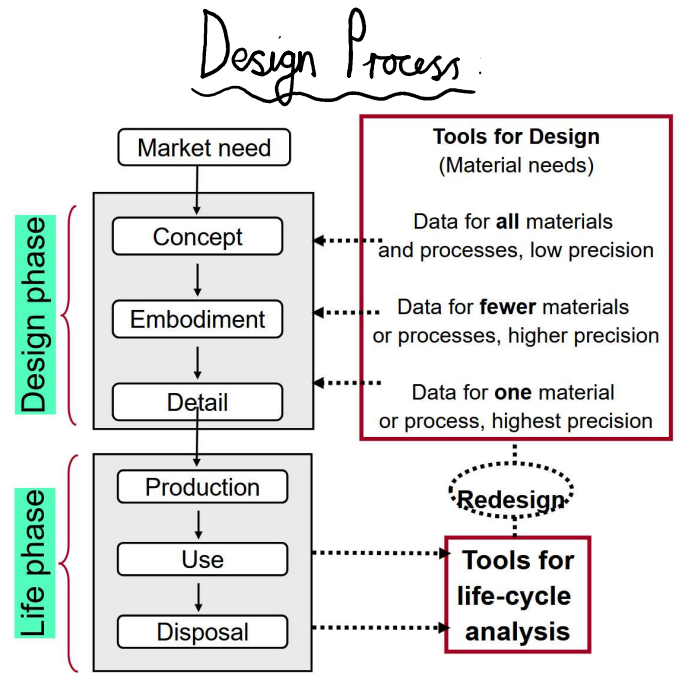
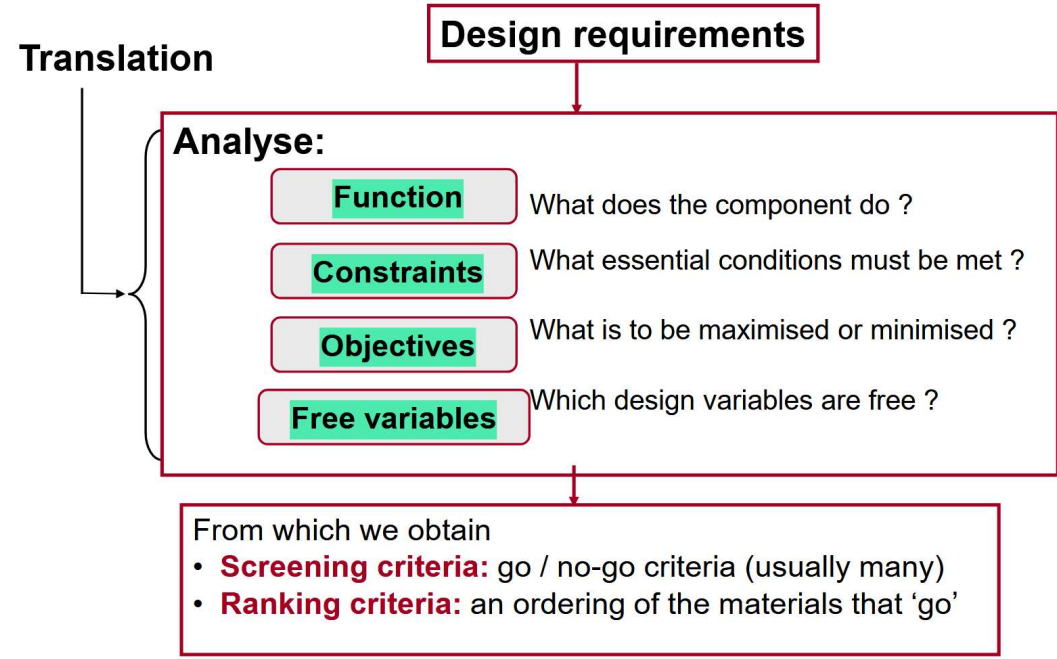




① → Translation of design requirements into a material specification.



② → Screening of materials that fail constraints.

- Using material charts
- E.g: Transportation systems, especially aerospace require high values of strength to weight ratio $\left(\frac{\sigma_f}{\rho}\right)$ or modulus to weight ratio $\left(\frac{E}{\rho}\right)$
- On modulus - density graph: Solids rather than porous materials. $(0.8-20 \text{ Mg m}^{-3})$

③ → Rank by ability to meet objectives; derive material index.

The steps:

- Identify function, constraints, objective and free variables. (list simple constraints for limit-stage).
- Write down equation for objective - the "performance equation".
- If it contains a free variable other than material identify the constraint that limits it.
- Use this to eliminate the free variable in performance equation.
- Read off the combination of material properties that maximizes the performance.

④ → Search supporting info for promising candidates.

Flywheel:

Function: Flywheel

Objective: • Max. energy per rotation $\left(\frac{E}{wt}\right)$

- Must not disintegrate

Constraints: • Reasonable Cost

- Adequate Fracture toughness

Free variables: • Material choice

Deriving material index:

Rotational Energy, $U = \frac{1}{2} I \omega^2$ $\left[I = \frac{\pi \rho R^4 t}{2}\right]$

Mass, $m = \rho V = \rho \times \pi R^2 t$

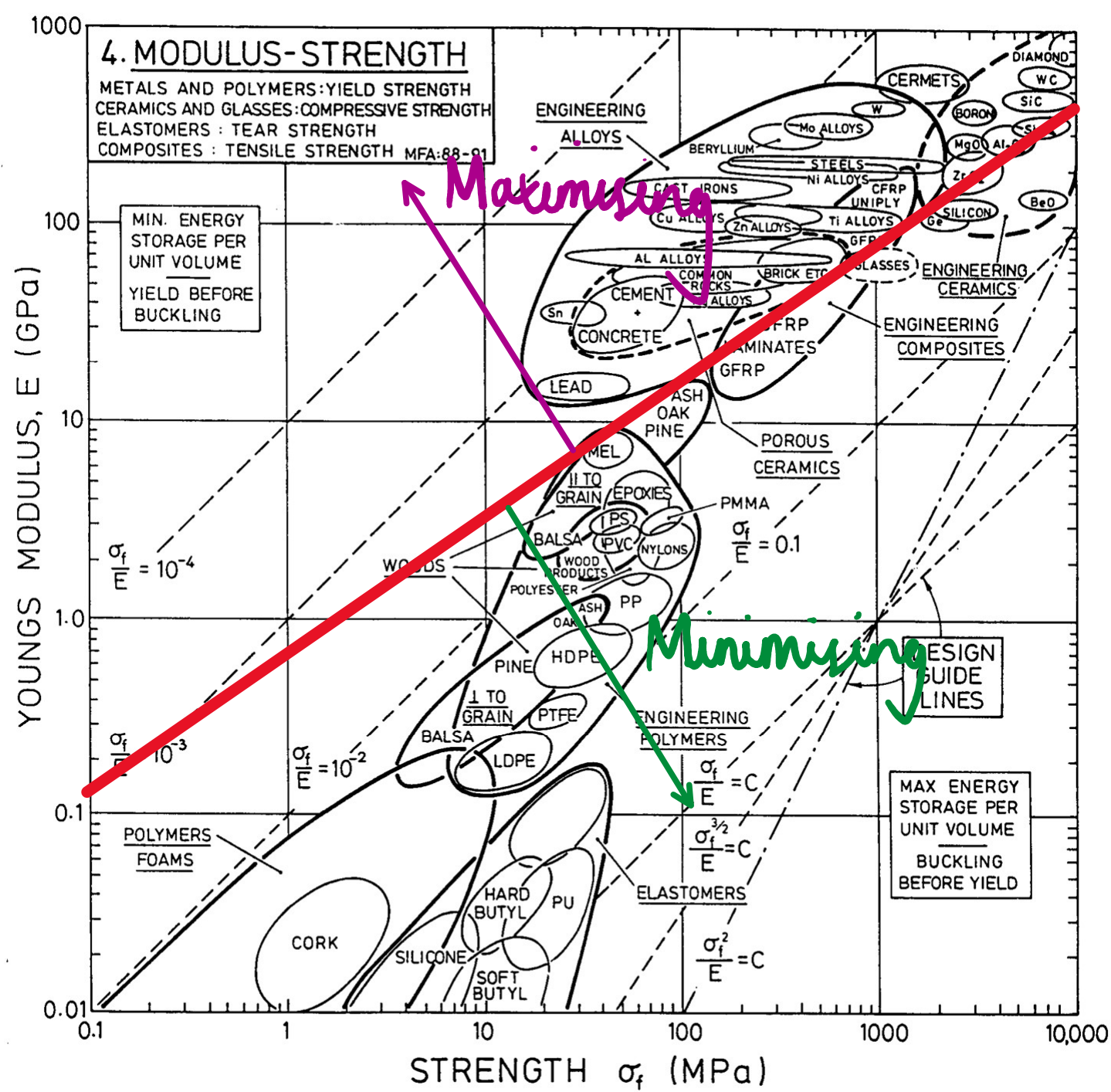
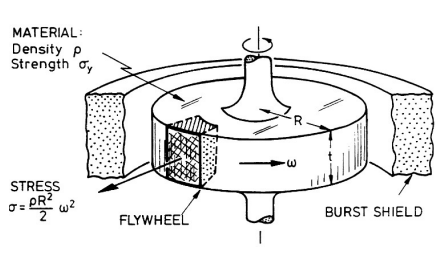
Energy $\frac{U}{m} = \frac{1}{2} \times \frac{\frac{1}{2} \pi \rho R^4 t \omega^2}{\pi R^2 t \rho} = \frac{1}{4} R^2 \omega^2$

Must not fail: $\sigma = \frac{(1+\nu)}{8} \rho R^2 \omega^2 < \sigma_f$

hence, $\frac{U}{m} \approx \frac{\sigma_f}{2\rho}$ (as $\frac{1+\nu}{8} \approx \frac{1}{2}$)

∴ To maximise: $\frac{U}{m}$

Need to maximise: $\frac{\sigma_f}{\rho}$



E.g.1:

Function: Light, strong Tie rod

Objective: minimise mass, $m = \rho V = \rho \times A L$

Constraints: • Specified length L

- Must not fail under load $F \Rightarrow \frac{F}{A} < \sigma_f$
- Adequate fracture toughness

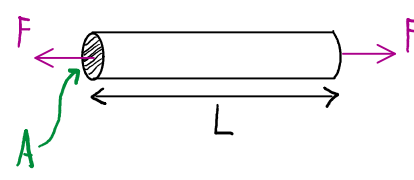
Free variables: • Material choice

• Section Area, A

Deriving material index: $A = \frac{F}{\sigma_f}$; $m = \rho \times A L$

$m = F L \left(\frac{\rho}{\sigma_f}\right)$

∴ To minimise mass (m), minimise $\left(\frac{\rho}{\sigma_f}\right)$ or maximise $\left(\frac{\sigma_f}{\rho}\right)$



E.g.2:

Function: Stiff, light beam

Objective: minimise mass, $m = \rho V = \rho \times b^2 L$

Constraints: • Specified length L

- Stiffness of beam, $S = \frac{C E I}{L^3}$
- Second moment of Area, $I = \frac{b^4}{12}$

Free variables: • Material choice

• Edge length, b

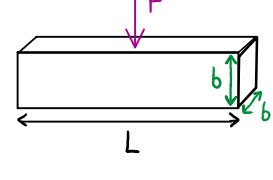
Deriving material index: $b^2 = \frac{m}{\rho L}$; $S = \frac{C E b^4}{12 L^3}$

$12 S L^3 = C E \left(\frac{m}{\rho L}\right)^2$

$m = \rho L \left(\frac{12 S L^3}{C E}\right)^{0.5}$

$m = \left(\frac{12 S L^6}{C}\right)^{0.5} \left(\frac{\rho}{E}\right)$

∴ To minimise mass (m), minimise $\left(\frac{\rho}{E^{\frac{1}{2}}}\right)$ or maximise $\left(\frac{E^{\frac{1}{2}}}{\rho}\right)$



E.g.3:

Function: Stiff, light panel

Objective: minimise mass, $m = \rho V = \rho \times \omega \times t \times L$

Constraints: • Specified length L

- Stiffness of panel, $S = \frac{C E I}{L^3}$
- Second moment of Area, $I = \frac{\omega t^3}{12}$

Free variables: • Material choice

• Panel thickness, t

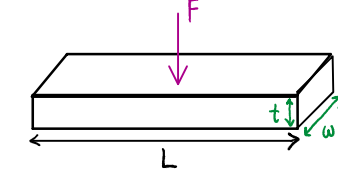
Deriving material index: $t = \frac{m}{\rho \omega L}$; $S = \frac{C E \omega t^3}{12 L^3}$

$12 S L^3 = C E \omega \left(\frac{m}{\rho \omega L}\right)^3$

$m = \rho L \omega \left(\frac{12 S L^3}{C E \omega}\right)^{\frac{1}{3}}$

$m = \left(\frac{12 S L^6}{C \omega}\right)^{\frac{1}{3}} \left(\frac{\rho}{E^{\frac{1}{3}}}\right)$

∴ To minimise mass (m), minimise $\left(\frac{\rho}{E^{\frac{1}{3}}}\right)$ or maximise $\left(\frac{E^{\frac{1}{3}}}{\rho}\right)$



- A lead flywheel can only store 3kJ/kg before it fails.
- Children's toys can still use lead for the flywheel, as the rate at which the disk is spun is quite small. Here even a Super-child could not spin the wheel fast enough to lead to failure.
- The rotational speed is now actually constrained by the driving mechanism. Here it is the maximum density that becomes the most suitable.
- These children's toy flywheels are very efficient when performance is velocity limited not strength limited. Other high-density materials such as gold and uranium are even better but are unsuitable through cost.

Materials	M (kJ/kg)	Comment
Ceramics	200-2000	Weak in tension
Composites - CFRP	100-500	The best choice
Composites - GFRP	100-400	Cheaper than CFRP and still good.
Beryllium	300	Expensive, but very good.
HS steels	100-200	Not bad and quite cheap
HS Al alloys	100-200	Not bad and cheap
HS Mg alloys	100-200	Not bad but more expensive
Ti Alloys	100-200	Not bad but even more expensive
Lead Alloys	3	Good for velocity limited performance not strength limited performance
Cast Iron	8-10	Good for velocity limited performance not strength limited performance